Muhammad Argya Vityasy

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Education

Universitas Gadjah Mada **Yogyakarta, Indonesia**
Bachelor of Computer Science | Cumulative GPA: 3.85/4.00 (113/144 credits) *August 2023 – Present*

- **Relevant Coursework:** Algorithms and Data Structures, Scalable Software Engineering, Software Engineering Methods, Cloud Computing, Computer Network, Cyber Security

Work Experience

Backend Developer | PT Satria Duta Raya **O**
B2B procurement platform enabling businesses to discover ranked products through an asynchronous recommendation pipeline.

- Designed and implemented a production-grade Go backend (Gin) with strict domain boundaries across authentication, buyer, supplier, product, order, and notification domains.
- Built an event-driven product ranking pipeline using Redis Streams, decoupling heavy computation from request handling while ensuring consistency through idempotent consumers.
- Implemented failure-aware background processing with Dead Letter Queues and bounded retries to safely handle transient errors.
- Integrated PostgreSQL and MinIO for secure, S3-compatible object storage, supporting transactional integrity and safe object handling.
- Operated the system as a live service with full observability, monitoring request throughput, latency, background worker health, and resource utilization using Prometheus and Grafana.

Backend Developer | Jogja International Model United Nation 2025 **April 2025 – O**
Annual academic simulation of the United Nations bringing together students worldwide for diplomatic debates.

- Architected and deployed a scalable event management platform using Go (Gin-Gonic), PostgreSQL, and AWS S3, leveraging Docker-based deployments and optimized database indexing and schema to process 10,000+ API requests, 500+ concurrent registrations, and 5GB+ of user data with high reliability.
- Enhanced operational efficiency by 40% by building a role-based admin dashboard that automated user verification and approval workflows, reducing manual processing time for 150+ staff actions and improving data integrity.
- Maintained 99.9% uptime under peak load of 500+ daily users through backend performance tuning, API integration optimization, and real-time monitoring for consistent, low-latency user experience.

Backend Developer | Organisasi Mahasiswa Ahli Teknologi Informasi UGM **March 2**
Student-led IT organization comprising 8 specialized divisions and 80+ members managing real-world software projects.

- Engineered a reusable, modular backend framework leveraging Go, Node.js, PostgreSQL, and Docker, reducing project development and deployment time by 60% and improving maintainability across multiple services.
- Deployed the framework in 4 production-grade client projects, handling 3,000+ user registrations, high-concurrency workloads, and maintaining 99.9% uptime under real-world traffic conditions.
- Led and mentored a 15-member backend engineering group, driving technical growth through weekly sessions on CI/CD pipelines, distributed systems, caching strategies, message brokers, Kubernetes orchestration, automated testing, and secure authentication practices.

Projects

Mock Test of Indonesia's National University Entrance Examination | GitHub

- Designed and deployed a four-service microservices backend for a national exam platform using Go (Gin) and Docker.
- Supported 200+ concurrent users with 99.9% uptime and sub-50 ms API latency under load.
- Implemented token-based authentication and a timed exam engine, exposing 20+ REST APIs.
- Processed 150+ exam sessions from a 1,000-question bank with ¡5% user error rate.
- Designed a migration plan to introduce Kafka-based asynchronous processing and Kubernetes orchestration for future scalability.

Additional Skills & Interests

Technical Skills: Git, GitHub, Gin-Gonic (Go), Express (TypeScript), PostgreSQL (SQL), MongoDB (No-SQL), Kafka, Redis, Loki, Grafana, Prometheus, Docker, Terraform, Ansible, Neovim

Interests: Cloud computing, DevSecOps, distributed systems, container orchestration, system design & scalability